

HR Analytics Dashboard Documentation

Project Title

HR Analytics Dashboard | Employee Attrition & Workforce Insights

1. Project Overview

The objective of this project was to analyze employee attrition patterns and workforce behavior using HR analytics techniques. The dashboard was designed to help HR teams and management understand the key factors contributing to employee turnover and workforce trends.

This project combines:

- SQL for data analysis
- Python for data cleaning and exploratory data analysis
- Excel for preprocessing
- Power BI for dashboard development and visualization

The final solution provides interactive dashboards with meaningful business insights related to employee attrition, salary trends, work-life balance, job satisfaction, promotion delays, and workforce engagement.

2. Business Problem

Employee attrition is one of the major challenges for organizations because it impacts:

- Recruitment costs
- Employee productivity
- Team stability
- Organizational performance

The goal of this project was to identify:

- Which employees are more likely to leave
 - Which departments are facing higher attrition
 - Whether salary, work-life balance, overtime, promotion delays, and job involvement impact attrition
 - Workforce patterns and employee engagement trends
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3. Dataset Information

Dataset Used: IBM HR Analytics Employee Attrition & Performance Dataset

Dataset Source: IBM Sample HR Dataset

Total Records: 1470 Employees

The dataset contains employee-related information such as:

- Age
 - Gender
 - Department
 - Job Role
 - Monthly Income
 - Attrition Status
 - Job Satisfaction
 - Work Life Balance
 - Business Travel
 - Years at Company
 - Years Since Last Promotion
 - Job Involvement
 - Overtime
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4. Data Cleaning & Preprocessing

The dataset was cleaned before analysis.

Cleaning Steps Performed

Removed Unnecessary Columns

The following columns were removed because they were not meaningful for business analysis:

- DailyRate
- HourlyRate
- MonthlyRate

Column Renaming

- EmployeeNumber column was renamed to EmployeeID

Data Validation

- Checked for null values
- Checked for duplicate rows
- Verified data types
- Validated categorical columns

Additional Transformations

Custom groups and categories were created for:

- Age Groups
 - Salary Slabs
 - Years Since Last Promotion Groups
 - Years at Company Groups
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5. Tools & Technologies Used

Tool	Purpose
SQL	Data querying and business analysis
Python	Data cleaning and EDA
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Excel	Initial preprocessing
Power BI	Dashboard creation and KPI analysis

6. SQL Analysis

SQL was used to extract insights from the HR dataset.

Analysis Performed

- Employee attrition analysis
- Department-wise attrition
- Job role analysis
- Salary analysis
- Overtime analysis
- Job satisfaction trends
- Workforce distribution analysis

Example SQL Insights

- Sales and Laboratory roles showed higher attrition.
 - Employees working overtime were more likely to leave.
 - Employees with lower salaries showed higher attrition.
 - Early-stage employees contributed significantly to employee turnover.
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7. Python Analysis

Python was used for Exploratory Data Analysis (EDA) and preprocessing.

Python Operations Performed

- Dataset inspection
- Statistical analysis
- Missing value analysis
- Duplicate checking
- Correlation analysis
- Distribution analysis
- Visualization of employee trends

Python Libraries Used

- Pandas
- NumPy
- Matplotlib

Visualizations Created

- Attrition distribution
 - Salary distribution
 - Age distribution
 - Job satisfaction analysis
 - Overtime analysis
 - Correlation heatmap
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8. Power BI Dashboard Development

A multi-page interactive dashboard was developed using Power BI.

Dashboard Features

- KPI cards
 - Interactive slicers
 - Attrition analysis
 - Salary analysis
 - Workforce insights
 - Professional dashboard design
 - Business-focused storytelling
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9. Dashboard Pages

Page 1 – Executive Overview

This page provides a high-level summary of workforce metrics.

KPIs Included

- Total Employees
- Attrition Count
- Attrition Rate
- Average Salary
- Average Age

Visualizations Included

- Attrition by Department
- Attrition by Age Group
- Overtime vs Attrition
- Attrition by Job Satisfaction

Key Insights

- Research & Development showed high employee attrition.
 - Employees doing overtime showed higher attrition.
 - Certain age groups had significantly higher employee turnover.
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Page 2 – Employee Analysis

This page focuses on employee behavior and attrition drivers.

Visualizations Included

- Attrition by Work-Life Balance
- Attrition by Business Travel
- Attrition by Years Since Last Promotion
- Attrition by Salary Slab

Key Insights

- Employees with lower salaries contributed highest attrition.
 - Promotion delays influenced employee turnover.
 - Work-life balance impacted employee retention.
 - Frequent business travel affected attrition patterns.
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Page 3 – Advanced Workforce Insights

This page focuses on deeper workforce trends and compensation analysis.

Visualizations Included

- Attrition by Job Role
- Average Salary by Job Role
- Attrition by Years at Company
- Attrition by Job Involvement

Key Insights

- Laboratory Technicians and Sales Executives showed high attrition.
 - Employees with lower job involvement had higher attrition rates.
 - Employees in the first few years at the company showed higher turnover.
 - Managers and Research Directors had the highest salaries.
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10. DAX Measures Used

Total Employees

```
Total Employees = COUNT('IBM HR Analytics'[EmployeeID])
```

Attrition Count

```
Attrition Count =  
CALCULATE(  
    COUNT('IBM HR Analytics'[Attrition]),  
    'IBM HR Analytics'[Attrition] = "Yes"  
)
```

Attrition Rate

```
Attrition Rate =  
DIVIDE(  
    [Attrition Count],  
    [Total Employees]  
)
```

Average Salary

```
Average Salary =  
AVERAGE('IBM HR Analytics'[MonthlyIncome])
```

11. Key Business Insights

Workforce Insights

- Employee attrition was concentrated in specific departments and job roles.
 - Employees with lower salaries showed significantly higher attrition.
 - Overtime employees had higher employee turnover.
 - Employees with low job involvement showed higher attrition rates.
 - Early-stage employees (0–2 years) contributed most to attrition.
 - Promotion delays impacted employee retention.
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12. Challenges Faced

During this project several challenges were addressed:

- Selecting meaningful business KPIs
 - Choosing appropriate visualizations
 - Managing dashboard layout and spacing
 - Creating custom groups and sorting logic in Power BI
 - Maintaining design consistency across pages
 - Building professional storytelling through dashboard pages
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13. Project Outcome

The final dashboard successfully provides:

- Workforce analysis
- Attrition monitoring
- Compensation insights
- Employee engagement analysis
- Interactive HR reporting

The dashboard can help HR teams and management make data-driven decisions related to employee retention and workforce planning.

14. Skills Demonstrated

This project demonstrates skills in:

- SQL Querying
 - Data Cleaning
 - Exploratory Data Analysis
 - Data Visualization
 - Dashboard Design
 - KPI Development
 - Business Intelligence
 - Power BI
 - Python Programming
 - HR Analytics
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15. Conclusion

The HR Analytics Dashboard project provided hands-on experience in end-to-end data analytics workflow including data cleaning, analysis, visualization, and business storytelling.

The project successfully transformed raw HR data into meaningful insights that can support better workforce planning and employee retention strategies.

This project also strengthened practical skills in SQL, Python, Power BI, and business analytics.

16. Future Improvements

Potential future enhancements include:

- Predictive attrition modeling using machine learning
 - Real-time dashboard integration
 - Drill-through analysis
 - Advanced Power BI tooltips and bookmarks
 - Automated reporting workflows
 - Cloud deployment and sharing
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17. Author

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Aspiring Data Analyst

Skills: SQL | Python | Excel | Power BI | Data Analytics | Data Visualization | Business Intelligence